Prologue

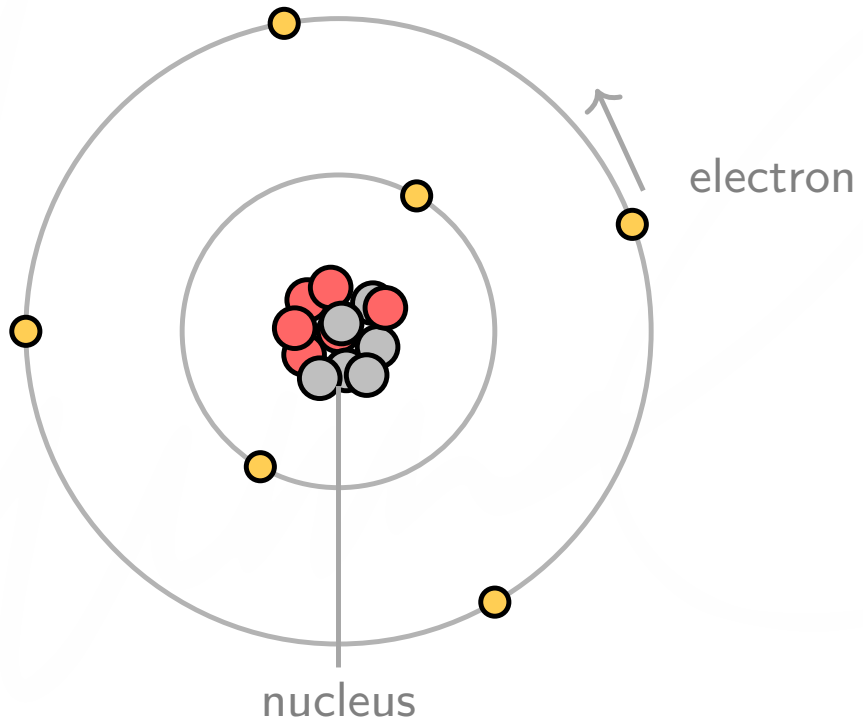
Computer Organization

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All matter – copper, air, you – is made of **atoms**: a small, dense **nucleus** surrounded by a cloud of **electrons**.



the planetary model of the atom

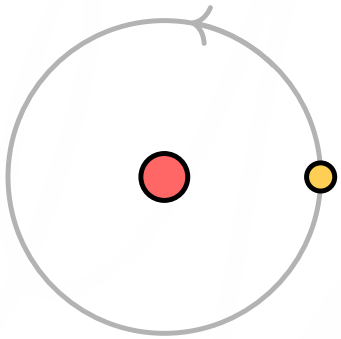
The nucleus and the electron cloud are built from three particles:

- **protons**:
  positively charged, sit in the nucleus
- **neutrons**:
  no charge, sit in the nucleus alongside the protons
- **electrons**:
  negatively charged, occupy the space around the nucleus

Opposite charges attract, so the negative electrons are held near the positive nucleus.

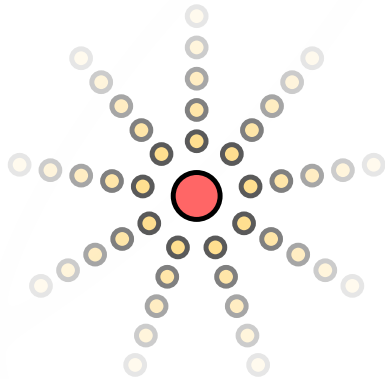
Our current understanding is that electrons are described by **probability clouds** around the nucleus, rather than following fixed orbits.

planetary model



electrons follow a fixed path

electron cloud

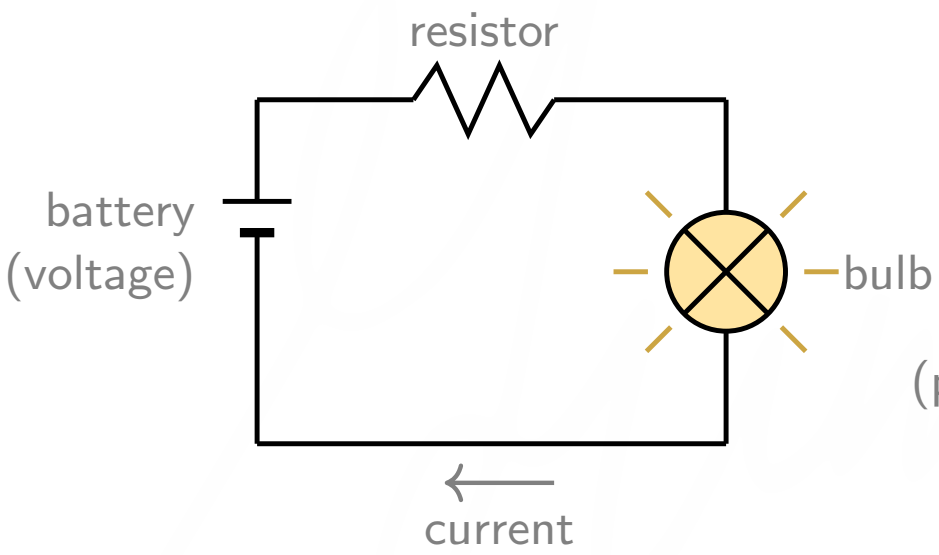


electrons exist as a probability cloud

Nevertheless, what matters for us: some electrons are **free** to move between atoms – that movement is **electric current**.

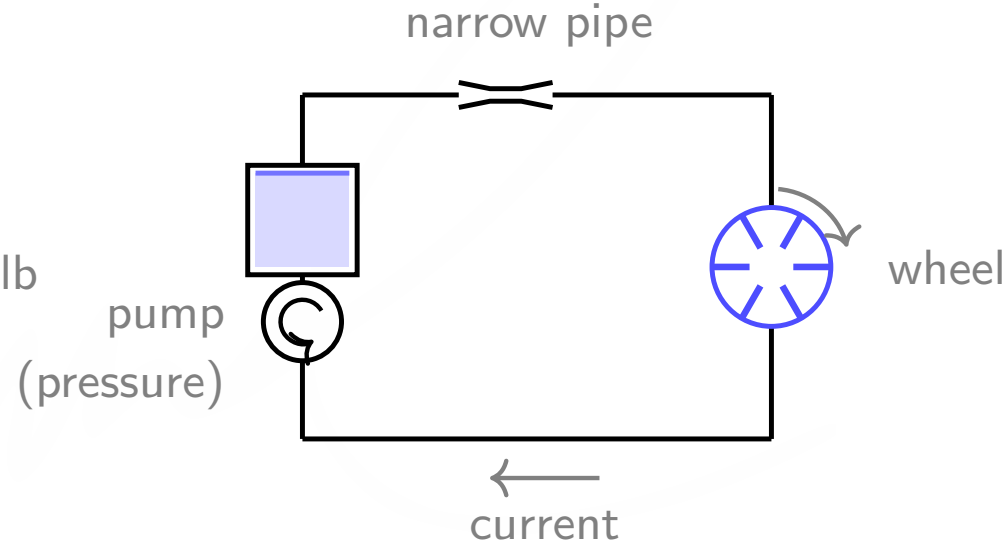
A computer is a machine for **representing** and **manipulating** information using electrical signals.

circuit



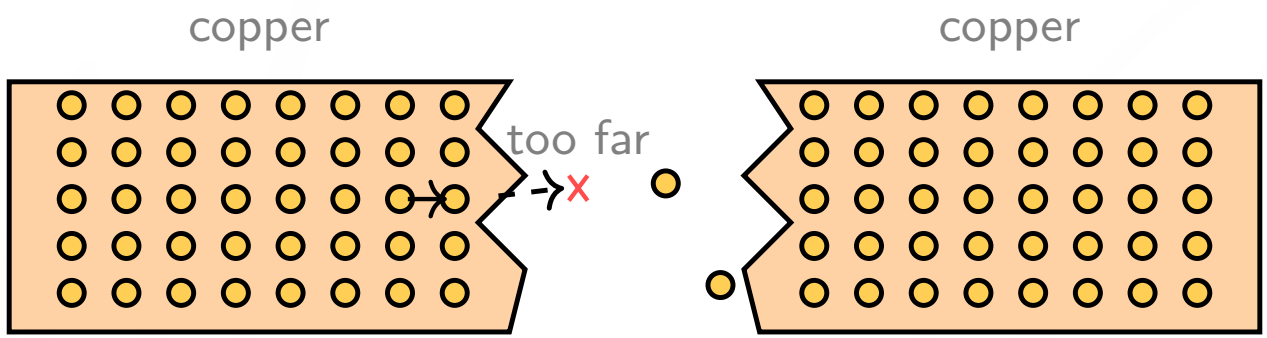
the battery pushes electrons through the resistor and bulb

water analogy



the pump pushes water through the narrow pipe and wheel

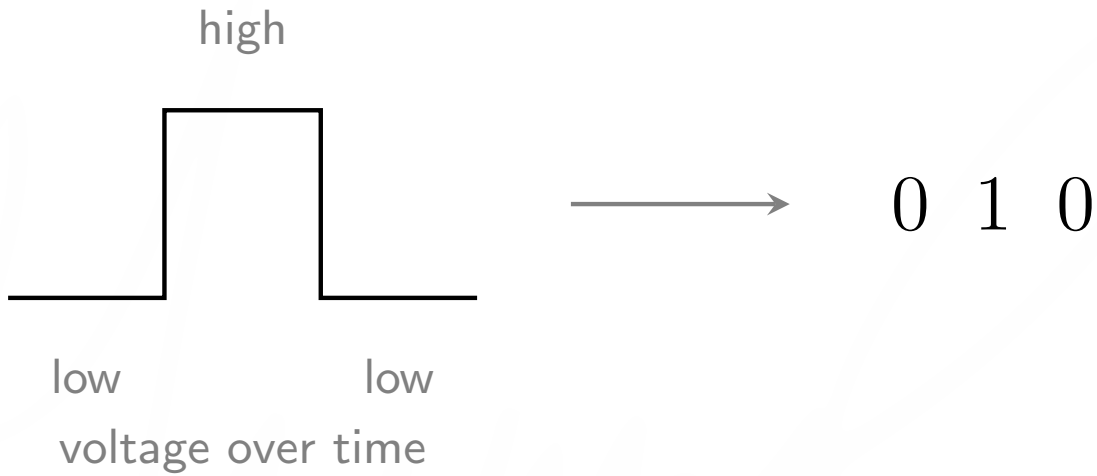
Inside the wire, electrons are close together, so the electrical disturbance can easily ripple from one electron to the next. In air, electrons are far apart, making it much harder for that ripple to continue[1].



with a voltage applied, each electron has enough energy to push its neighbor in the copper wire, but not across the gap

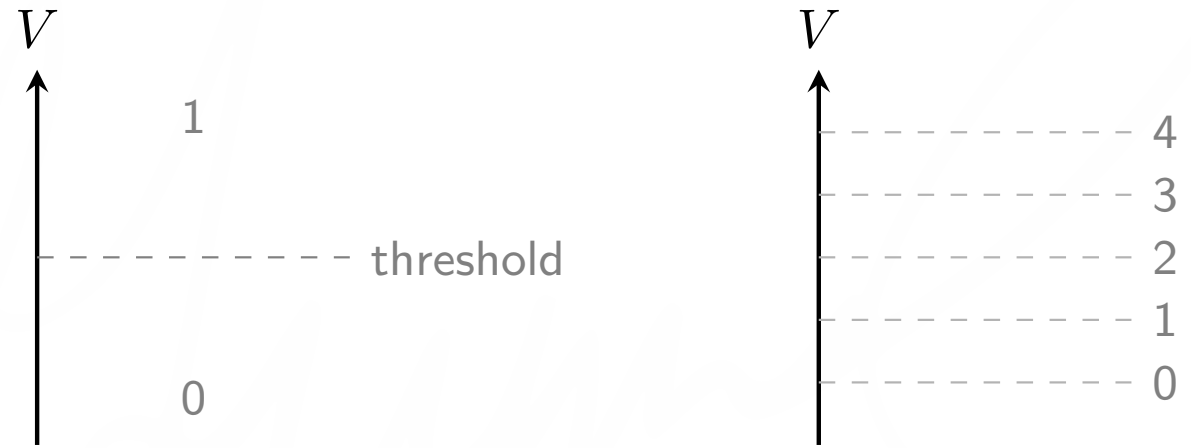
[1] More precisely, metals like copper contain many *free* electrons that respond to the electric field, whereas air contains very few free electrons under normal conditions.

We break up the voltage range into two levels.



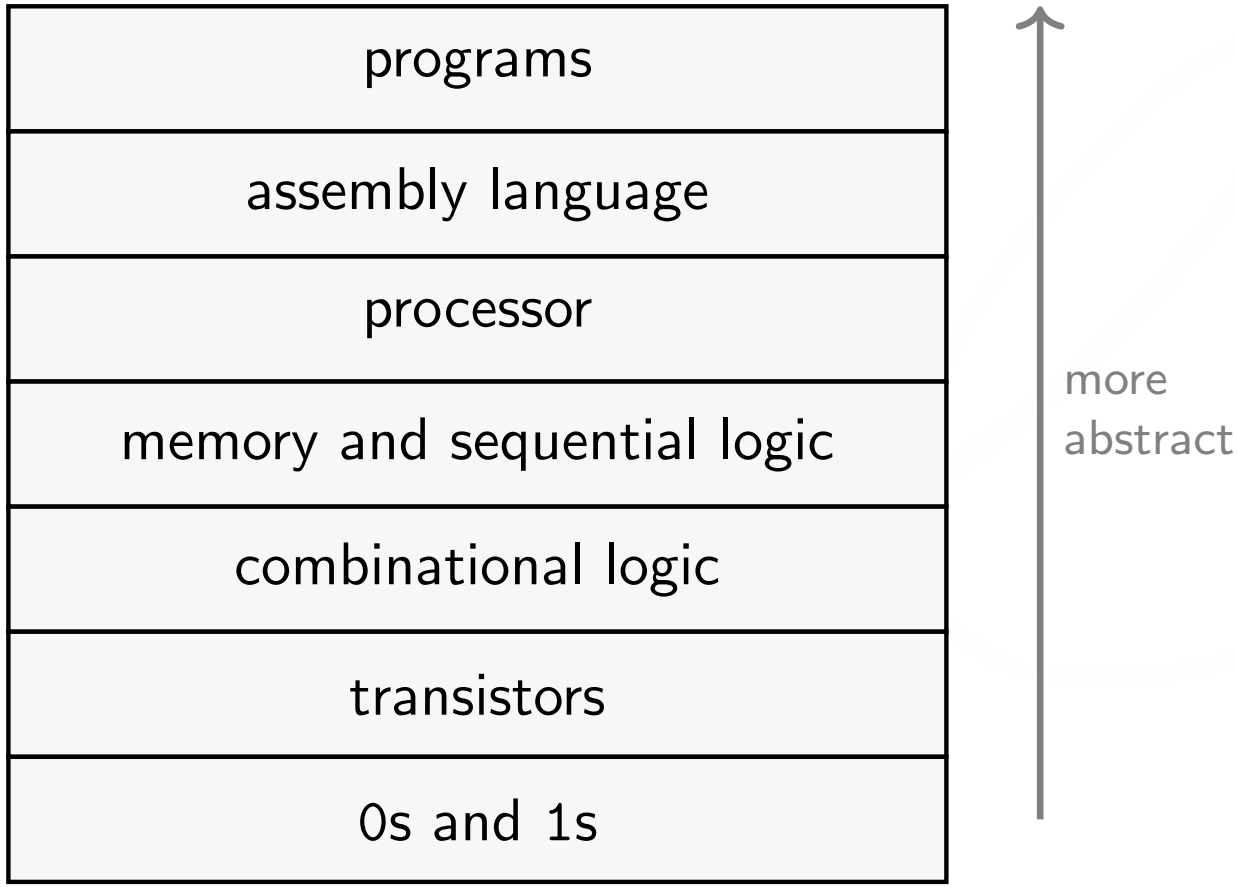
a voltage above some upper threshold is read as a 1, and a voltage below a lower threshold is read as a 0

We could break up the voltage range into more than two levels, but we don't due to **signal noise**.



fewer levels means that each level can be more reliably read

Starting with these binary signals (0s and 1s), we study how to build a computer layer by layer. Each layer hides the details of the one below it.



combinational logic

We begin with circuits whose output depends only on their current input.

- **logic gates**:
  the basic operations on binary values
- **circuit design**:
  combining gates to compute any function
- **arithmetic**:
  representing numbers and adding them in hardware

sequential logic

Next we add memory, so that output can depend on the past as well as the present:

- **flip-flops**:
  circuits that store a single bit
- **registers and counters**:
  groups of bits that hold and update values
- **state machines**:
  circuits that move through a sequence of states

architecture

Finally we assemble these parts into a working computer:

- **memory**:
  storing many values and reading them by address
- **the processor**:
  a datapath and control that carry out instructions
- **assembly language**:
  the instructions a program gives the processor